

Maulana Azad National Institute of Technology, Bhopal

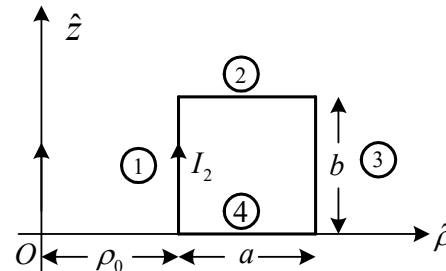
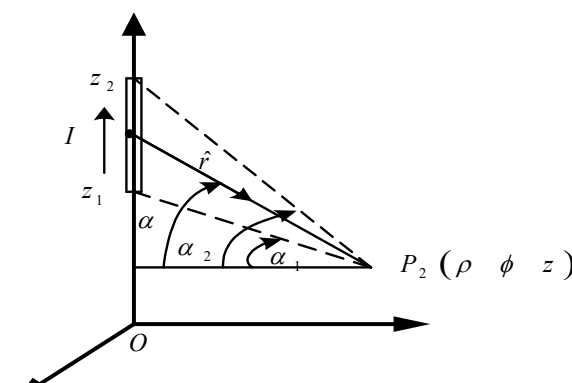
Department of Electrical Engineering

Class: B.Tech III Sem.

Sub: EMF& Materials (E-211)

Session- July-Nov. 2024

Assignments-4

Q. No.	Question
1	An infinity long coaxial conductor having inner conductor radius is 'a' and outer conductor has inner radius 'b' and thickness 't'. Inner conductor carrying current I and outer carrying the current $-I$, Find H every where. Assume that currents are uniformly distributed on both the conductors Also plot H vs ρ .
2	A filamentary current of 10 A is directed in from infinity to the origin on the positive X-axis, and then back out to infinite along the positive y-axis. Use Biot-Savart law to find H at $P(0 \ 0 \ 1)$.
3.	A rectangular loop carrying current I_2 is placed parallel to an infinity long filamentary wire carrying current I_1, as shown in Fig. Show that the force experienced by the loop is given by $F = -\frac{\mu_0 I_1 I_2 b}{2\pi} \left[\frac{1}{\rho_0} - \frac{1}{\rho_0 + a} \right] \hat{\rho}$ 
4	A negative charge $Q = -40 \text{ nC}$, is moving with a velocity of $6 \times 10^6 \text{ m/sec}$. in a direction specified by the unit vector $\hat{a} = -0.48\hat{x} - 0.6\hat{y} + 0.64\hat{z}$. Find the magnitude of the vector force exerted on the moving particle by the field (a) $B = 2\hat{x} - 3\hat{y} + 5\hat{z}, \text{ mT}$ (b) $E = 2\hat{x} - 3\hat{y} + 5\hat{z}, \text{ kV/m}$ (c) B & E acting together.
5	For the finite length of current on the z-axis as shown in Fig. Using Biot-Savart law show that $H = \frac{I}{4\pi\rho} [\sin \alpha_2 - \sin \alpha_1] \hat{\phi}$ Also prove that $H = \frac{I}{4\pi\rho} [\cos \alpha_2 - \cos \alpha_1]$ 

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